

Question Paper

Exam Date & Time: 11-Jun-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

SECOND SEMESTER B.TECH. EXAMINATIONS - JUNE 2025
SUBJECT: BIO 1071/BIO_1071 - BIOLOGY FOR ENGINEERS
(MAKE-UP)

Marks: 50

Duration: 180 mins.

Answer all the questions.

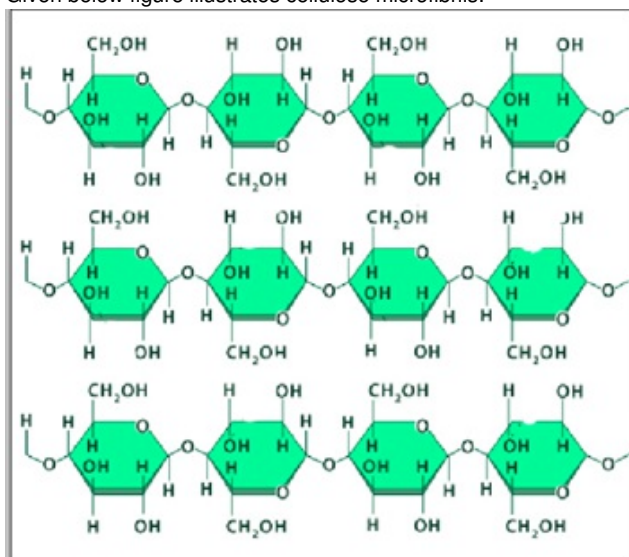
- 1A) Alien life forms are most probably based on silicon. Justify. (3)

Course Outcome: "CO1" / Cognitive Level: "Application"

- 1B) Proteins can form four levels of structures, while carbohydrates not. Justify. (3)

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- 1C) Given below figure illustrates cellulose microfibrils: (4)



- i) Copy the figure and illustrate how hydrogen bonding can form in these microfibrils.
ii) How the linkage of cellulose is different from amylose?
iii) Humans can digest amylose but not cellulose. Why?
(2+1+1 = 4 mark)

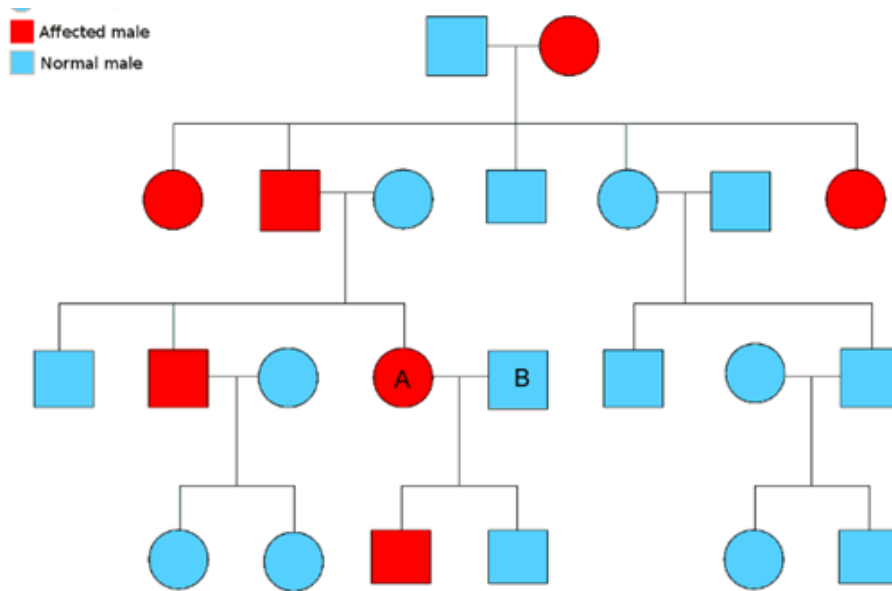
Course Outcome: "CO1" / Cognitive Level: "Understand"

- 2A) In a dihybrid cross between pea plants, you studied the inheritance of seed shape (round, R; wrinkled, r) and seed color (yellow, Y; green, y). You crossed plants heterozygous for both traits (RrYy) with each other. The following phenotypic ratio was obtained - You got a ratio of Round Yellow : Wrinkled green : Round green : Wrinkled yellow 9:1:3:3
a) Construct a Punnett square to represent this cross.
b) What you can predict from the results?
c) What are the possible genotypes of Wrinkled yellow?
(1+1+1 = 3 marks)

Course Outcome: "CO2" / Cognitive Level: "Application"

- 2B) Analyse the following pedigree chart: (3)

● Affected female
● Normal female



- Predict the best possible mode of inheritance? Justify?
 - Predict the genotype of elderly male and female.
 - Calculate the probability of the third child being expressing the character for couples A and B.
- (1+1+1 = 3 marks)

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- 2C) A scientist is studying two genes, A and B, in a plant species. Gene A controls flower color (red or white), and gene B controls leaf shape (rounded or jagged). The dominant allele for gene A is represented as A (red), and the recessive allele as a (white). Similarly, the dominant allele for gene B is represented as B (rounded), and the recessive allele as b (jagged). (4)

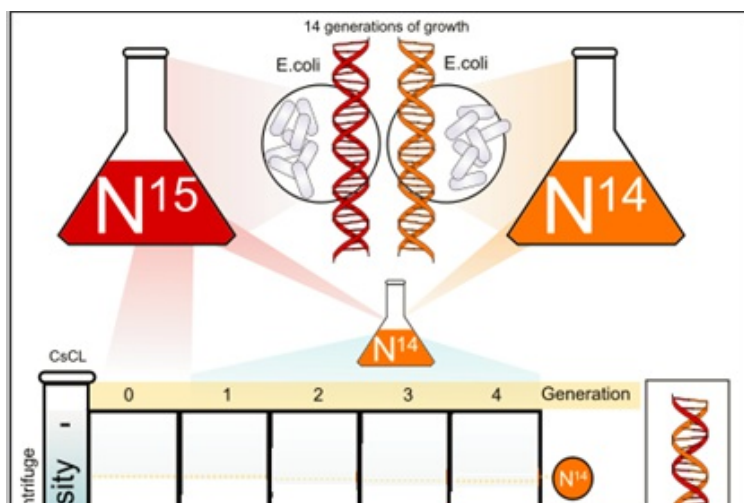
To study the inheritance patterns of these traits, the scientist performs a dihybrid test cross between a plant heterozygous for both traits (AaBb) and a plant homozygous recessive for both traits (aabb). The offspring display the following phenotypes and numbers:

- Red flowers, rounded leaves: 40
- Red flowers, jagged leaves: 10
- White flowers, rounded leaves: 12
- White flowers, jagged leaves: 38

- Exhibit the cross in the form of a Punnett square.
- Identify the parental and recombinant phenotypes.
- Calculate the recombination frequency for these genes (factors).
- Based on your calculations, determine whether these genes are likely linked or assort independently.

Course Outcome: "CO2" / Cognitive Level: "Understand"

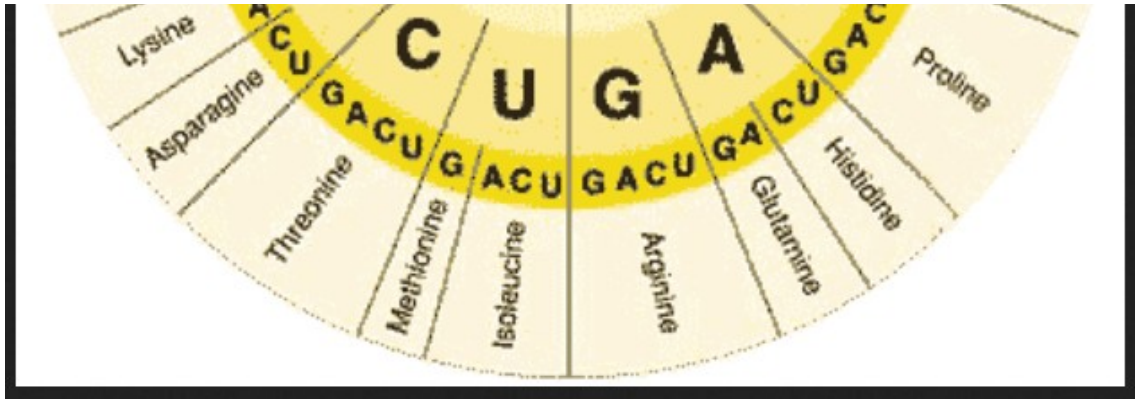
- 3A) Predict how the bands will appear in density gradient centrifugation for (a) Copy the figure and mark the bands conservative and (b) Copy the figure and mark the bands semi conservative mode of replication. (3)





(3)

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- A circular genetic code chart (codon wheel) showing the mapping of mRNA codons to amino acids. The center contains the four bases: G, U, A, C. The first ring shows the second base, and the outer ring shows the third base. Amino acids are labeled around the perimeter, including Glycine, Phenylalanine, Leucine, Serine, Tyrosine, Stop, Cysteine, Stop, Tryptophan, Leucine, Alanine, Valine, Arginine, and Serine.



Course Outcome: "CO3" / Cognitive Level: "Analysis"

- 3C) How mRNA is structurally organized to serve its function? Justify with the help of figure (4)

Course Outcome: "CO3" / Cognitive Level: "Evaluate"

- 4A) Two populations of the same bird species are separated by a mountain range. Over time, the populations of birds develop different beak shapes. Suggest how this could have happened. (3)

Course Outcome: "CO4" / Cognitive Level: "Understand"

- 4B) Explain the fossil evidences of evolution (3)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- 4C) Justify how mutations contribute to evolution (4)

Course Outcome: "CO4" / Cognitive Level: "Understand"

- 5A) i) How do vaccines work? (3)
ii) How long does immunity last after getting a vaccine?
iii) With adequate levels of hygiene, sanitation and clean water, is there still a need for vaccination?

Course Outcome: "CO5" / Cognitive Level: "Understand"

- 5B) Explain the biomimetic principles from the following illustrations (3)

1.



2.

(1½+1½ = 3 marks)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

5C)

Briefly explain the bioinspiration that may be developed from the following:

(4)

- i) Burrs of plants
- ii) Shark skin
- iii) Gecko
- iv) Bird's beak

Course Outcome: "CO5" / Cognitive Level: "Understand"

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Event Name			Subject
MIT MPL and BLR- BTech-MSc-MCA - II-IV and VI Sem - Makeup Examination - Jun 2025			BIO 1071-CHM:BIOLOGY FOR ENGINEERS
#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	10
Total			50

* NA - CO Not Selected

Event Name			Subject
MIT MPL and BLR- BTech-MSc-MCA - II-IV and VI Sem - Makeup Examination - Jun 2025			BIO 1071-CHM:BIOLOGY FOR ENGINEERS
#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	6	22(44%)
3	Application	3	9(18%)
4	Analysis	5	15(30%)
5	Create	1	4(8%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)